

Sparse Codes Make Linear Attention Trustworthy: A Graph-Structured View of BDH-GPU's Attention Mechanism

Concepts: Linear Attention · Sparse Non-Negative Activations · Local Neural Computation, on graph-structured data · connected to BDH-GPU (Dragon Hatchling)

The design pressure

Standard softmax attention compares every pair of tokens directly, giving robust retrieval at the cost of a key-value cache that grows with context and $O(n)$ work per new query against everything stored so far. Linear attention removes the softmax nonlinearity and replaces pairwise comparison with an incrementally updated, fixed-size matrix — a fast-weight associative memory queried in $O(1)$ per token (Katharopoulos et al., 2020). The saving is real, but so is the cost: compressing many stored associations into one fixed-size matrix means they begin to interfere with each other, a capacity limit long known for linear associative memories. BDH-GPU (Kosowski et al., 2025) takes a specific, unusual position on this trade-off: rather than compressing keys into a small hidden dimension before applying linear attention, it runs linear attention directly in a very large neuron dimension n , forcing activations positive via ReLU, and reports them empirically sparse at roughly 5% active (Kosowski et al., 2025, Sec. 4.1).

What changes, and the trade-off it introduces

The mechanistic difference is where crosstalk gets controlled. Dense, low-dimensional linear attention accepts interference risk in exchange for a compact state. BDH-GPU instead expands into a high-dimensional but sparse code, so different stored associations are more likely to land on largely disjoint sets of active units — the same argument used to explain capacity in sparse, high-dimensional associative memories, echoing sparse coding accounts of the visual cortex (Olshausen and Field, 1997, cited in Kosowski et al., 2025). The cost is a larger state footprint per neuron, and a sparsity level that must land in a workable range: too dense and the scheme reduces to the interference-prone case; too sparse and there is not enough active capacity to represent much of anything.

Representative architectures, compared

	Softmax attention (Transformer)	Linear attention, low-dim (Katharopoulos et al., 2020 lineage)	BDH-GPU (Kosowski et al., 2025)
Per-token cost	$O(\text{context length})$	$O(1)$ after $O(1)$ update	$O(1)$ after $O(1)$ update
State	Raw KV-cache (grows with context)	Fixed small dense matrix	Fixed large sparse (~5% active) matrix
Interference risk	None (recomputed from raw keys)	High at scale (shown in our toy test)	Low at same scale (shown in our toy test)
Interpretability	Post-hoc only	Post-hoc only	Reported monosemantic synapses (Sec. 6.3)
Evidence status	Industry standard	Established literature	Single-group reported (see below)

Evidence, labeled correctly

BDH-GPU's headline result — matching GPT2-class Transformer scaling laws from 10M to 1B parameters on language and translation tasks — is a developer-reported result in the primary paper (Kosowski et al., 2025, Sec. 4.2), not yet an independent third-party reproduction. Its widely cited 97.4% Sudoku Extreme result is explicitly flagged, including in Pathway's own open-source repository, as coming from an internal implementation distinct from the released code, and is not independently reproduced. On the specific sparsity-versus-interference link this summary is about, our own small synthetic retrieval experiment (not a BDH reproduction) shows retrieval accuracy falling from about 100% to about 5% as memory load exceeds the embedding dimension for a dense linear-attention code, while a sparse (~5% active) code stays above 95% accuracy at the same load — a controlled illustration of the mechanism, not a claim about trained BDH models.

Roles of BDH and BDH-CQ

BDH-GPU is the direct, substantive anchor for this concept: its attention state is described by the primary paper itself as functioning like an associative memory ("like KV-cache, but organized differently," Sec. 4.1), built and queried exactly as a fast-weight linear-attention mechanism. BDH-CQ, a later system in the same family focused on gradient-free adaptation from demonstrations, has no direct mechanistic role in the linear-attention/sparsity story specifically — we note this rather than inventing a connection the source material does not support.

The most important open question

Our demonstration uses synthetic, independent random codes, for which near-orthogonality is guaranteed by construction. Whether the capacity benefit survives when codes emerge from training on real, correlated data — where keys are not independent and random — is not established here and is, to our knowledge, not fully settled by BDH-GPU's own results either. A second gap: no outside group has, to our knowledge, independently reproduced BDH-GPU's headline scaling-law or Sudoku findings, so confidence in the practical performance claims currently rests on a single research group's own reporting.

Primary sources: Kosowski, Uznanski, Chorowski, Stamirowska, Bartoszkiewicz, "The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain," arXiv:2509.26507 (2025) · Katharopoulos et al., "Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention," ICML 2020 · Haziza et al. (2025) · You et al., "Spark Transformer" (2025) · Buckman et al. (2024).